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Building an interactive visualization tool for athletes' performance data

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Sammanfattning

Sportvärlden blir mer och mer intresserad av att tillämpa dataanalys. Användandet av sensorer för att få information om händelser och rörelser i olika riktningar genererar stora datamängder, vilka till synes kan vara svåra att dra några direkta slutsatser av. Visualiseringar kan användas för att underlätta analysprocessen och möjliggöra mer avancerade insikter. Frågan är dock hur dessa visualiseringar skall implementeras samt vilka möjligheter som finns med detta. Denna rapport beskriver ett projekt för att designa ett interaktivt visualiseringsverktyg för olympisk kanot. Befintlig forskning och teorier kring informationsvisualisering kombinerades med relaterad forskning inom visuell dataanalys för sport och tillämpades i projektet. Arbetsprocessen följde en problemdriven metod under en kontinuerlig period av återkoppling med ämnesexperter. Kvalitativ data insamlades genom feedback-aktiviteter under en iterativ utvecklingsprocess av ett koncept för ett webbaserat analysverktyg. Resultaten antyder att visualiseringar understödjer och möjliggör skapandet av insikter på ett intuitivt sätt. Sammanfattningsvis kan detta koncept förbättra dataanalys, genom att göra den mer sofistikerad och samtidigt mer tillgänglig. Det kan även förbättra samarbetet mellan olika användarroller i analysprocessen. Överförbarheten till andra sporter påverkades dock av behovet av att designa verktyget specifikt för kanot. Därtill medförde multi-dimensionaliteten hos det aktuella datasetet ett antal utmaningar att ta hänsyn till i framtida forsknings- och utvecklingsarbete. Ett antal förbättringar blev föreslagna för konceptet och det bedömdes vara lovande för framtida forskning, utveckling och implementation, vilket också bör göras i samarbete med ämnesexperter.

Building an interactive visualization tool for athletes' performance data

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ABSTRACT

The world of competitive sports is becoming more and more interested in data analysis. Using sensors to track actions and trajectories produces large datasets that could be hard to draw conclusions from at face value. Visualizations can be used to make the process of data analysis easier and possibly allow for more advanced insights. The question is how visualizations should be implemented and what the possibilities are in doing so. Described in this paper is a project for designing an interactive visualization tool for olympic kayaking. Available research and theories of information visualization was combined with related research regarding visual data analysis for sports, and applied to the project. The process followed a problem-driven approach during a period of continuous feedback with domain experts. Qualitative data was collected throughout feedback sessions in an iterative process of developing a concept for a web-based tool. The results indicate that visualizations indeed do support and enable intuitive insight generation. In summary, the concept could improve data analysis, both making it more sophisticated and more accessible. Furthermore, it could improve collaborations between different user roles. However, needing to tailor the concept to a specific sport may affect the transferability to other sports, and the multidimensionality of the dataset produced a number of challenges to consider in future work. A number of improvements were suggested for the concept, and it was deemed promising for future research, development and implementation, which should be done in collaboration with domain experts.

1. INTRODUCTION

Sports today is becoming increasingly data-driven, with emphasis on using data analysis to enhance performance and ultimately reach higher athletic achievements [1]. Visualizations are an important tool for better understanding and gaining new insights from data, rather than just reading numbers at face value [11]. Using interactive visualizations, where the user can shape their own discovery process, is a way to improve contextualization and insight generation from various types of data [13]. However, there are many ways to design interactive visualization systems, and this must be done both appropriately and with the end-users in mind. Choosing the wrong visualizations may result in misrepresentation of the data, and end users subsequently drawing false conclusions [7]. This degree project is focused on these issues.

The topic of information visualization for sports/athletic data is an active research area. Much of the research is focused on team sports, but work exists for individual sports as well. Furthermore, much of the available research is focused around data from competitions.

1.1 Case description

The Swedish Olympic Academy (SOA) wants to implement an interactive visualization tool for multi-dimensional training data. Therefore, the purpose of this degree project is to explore the prerequisites for such a tool, that will help athletes, trainers, coaches, performance analysts etc. perform multivariable analysis. The interactive nature of the tool will allow

users to tailor the visualization to fit their individual needs, possibly gaining new insights from the training data. The tool is designed for *kayaking*, since the SOA had data available for this sport as part of an ongoing project about data collection and analysis for athletes. Currently, there is little available research in data visualization for kayaking.

The process followed in this project was inspired by the problem-driven approach proposed by Sedlmair, Meyer and Munzner in *Design study methodology* [14]. The practical result was a high-fidelity prototype: a web-application with interactive visualizations intended for further development and implementation by the SOA. Furthermore, this report explores opportunities and challenges with building an interactive visualization tool for multidimensional data, along with implications and insights for further implementation of the tool.

Thus, the following research question was formulated:

1.2 Research question

What are the opportunities and challenges with adaptive analysis of multidimensional datasets using interactive visualizations for athletes' performance data?

1.3 Delimitations

There are many ways to create interactive visualizations. However, the prototype developed in this degree project is in the form of a web-application, as requested by the SOA. Furthermore, while much research on this topic is centered around visualizing data from competitions, this project focused mainly on training data. While this project focused on kayaking, it is possible that the concept can be used in similar projects for other sports in the future.

2. THEORY AND RELATED WORK

This section establishes a theoretical and contextual background for this project.

2.1 The Swedish Olympic Academy

The work performed in this degree project was done in collaboration with the Swedish Olympic Academy (SOA). The SOA was founded in 2017 by the Swedish Olympic Committee (SOC), the Royal Institute of Technology (KTH), Karolinska Institutet (KI) and the Stockholm School of Economics (Handelshögskolan). The purpose of the foundation is to contribute to individual athletes reaching their full potential, the improvement of athletes' basic conditions, and strengthening the competitiveness of Swedish sports. The SOA has an ongoing project regarding training data for athletes, where data is recorded in real time by sensors while training.

2.2 Information visualization

Humans have the ability to view and reason visually about data, thereby gaining insight beyond simple data transfer. Simple insights could be summaries such as a maximum or average value of a certain dataset, while more complex insights involve finding patterns, relationships, structures, outliers and comparisons, for instance. Visualization is achieved by visual mapping, representing information structures, using visual overview strategies, navigation strategies and interaction techniques, and is a powerful way of aiding insight generation [10]. The concept of visualization takes place in the human mind, as a cognitive activity carried by external visual representations [8]. Additionally, the use of scientific visualization can reveal patterns, clusters, gaps, or outliers in statistical data, for instance [16].

2.3 Interactive visualizations

Visualizations mediate the interaction between a human and information [10]. An information visualization system appears to be at its most useful when end users can modify the input data, change the visual mapping, or manipulate the generated view [8]. There are many visual design guidelines describing how to achieve this in the best possible way. It has been summarized as what is known as the *Visual Information-Seeking Mantra*: overview first, zoom and filter, then details-on-demand [16]. Two types of

interactive visual representations can be defined: 1) *manipulable*, where a user can intervene by scrolling, switching between details and overview, and using focusing tools; 2) *transformable*, where the user can manipulate the preprocessing of data and/or the visual mapping itself [8].

Using visual elements when performing exploratory, information seeking tasks is far more efficient than using traditional text-based approaches [16]. Furthermore, during exploratory phases of research work, using interactive visualizations has been found to be effective for generating contextual insights [13]. Visualization systems have been described to have these four roles: 1) visually represent data to enhance data analytics; 2) visually display users' mental models, interpretations of the data, ideas, hypotheses and insight; 3) improve users' mental models by finding supporting and contradictory evidence for hypotheses; 4) let users organize and share ideas [17]. There is a growing interest for interactive exploration of sports data [1].

2.4 Interactive visualization tools

Humans need some sort of tool to facilitate visualization and interaction. If visualizations are to be effective, some approaches will need to utilize novel data structures, high-resolution color displays, fast data retrieval, specialized data structures and parallel computing [16]. However, it is important that the tools for this are designed with care, since choosing the wrong visualization might cause users to draw false conclusions [7]. Furthermore, a tool must be designed in such a way that it addresses the right problems, shows the right things, that the interaction actually works, and that the developed product executes fast enough [9]. The efficacy of an information visualization tool can be measured in various ways, but two aspects to look into could be its usage (documented through observations, interviews, surveys, logging, etc.), and expert users' success in achieving professional goals [15].

2.5 Multidimensional datasets

Relational and statistical databases can be manipulated as multidimensional data, meaning that items with n attributes become points in an n -dimensional

space. There are various ways in which to visualize this using 2D or 3D plots, but using the latter may be associated with disorientation and/or occlusion [16]. Early on in the design process, the data must undergo specification and selection: Is it continuous or discrete? Is there an inherent order? Is the domain spatial or temporal? What is the dimensionality of the data? What data must be shown, and what data is redundant? The answers to these questions will greatly influence the design of the visualization system [7].

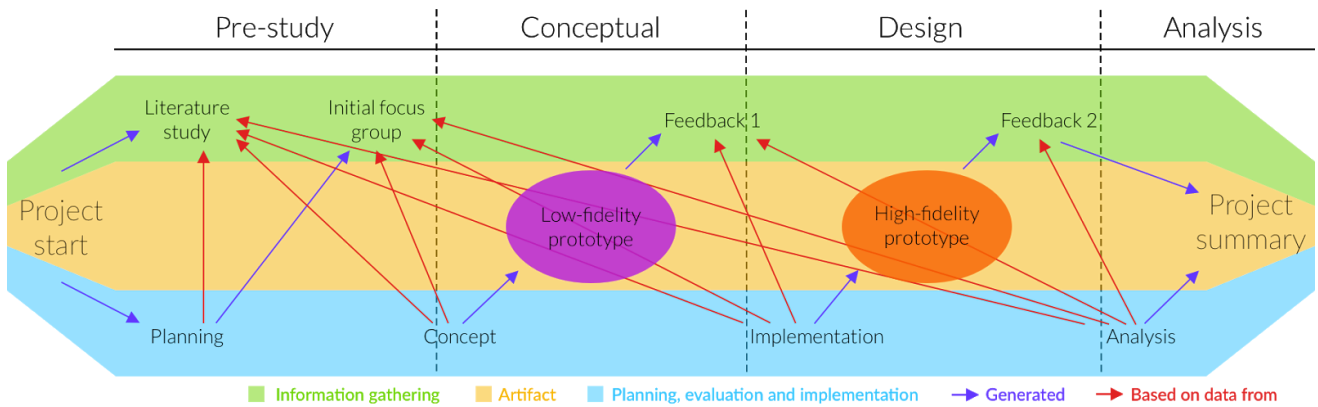
2.6 Adaptive analysis

Data analysis is mostly adaptive or exploratory in practice, meaning that previous conclusions and insights will inform future analysis [4]. A user's interpretation of data will also depend on their own prior knowledge about the topic in question. It is beneficial for users to navigate data based on this prior experience, when trying to extend their knowledge [12]. Since visualization is a cognitive activity, this will affect how the user interacts with the visualization tool and will impact the insight generation as a result.

2.7 Athlete's performance data

Sports is associated with many different types of data. The highly sport-specific *box score data* exists mainly to account for things such as points scored or finishing times. *Meta data* is often similar across sports, such as height and weight of an athlete. *Tracking data* has to do with in-game or in-race actions and trajectories. While this data could be applicable across sports, its relevance may vary [11]. Wearable sensors that record tracking data are applicable for many situations, and are used by sports organizations for various purposes, such as preventing injury or enhancing performance.

Because there is a wide range of types of sports data, there are many interesting challenges for visualization in this area [1]. For instance, a sport like fencing will rely heavily on tactics, which could make it difficult to discover hidden patterns using traditional statistical analysis [18]. Furthermore, there are many types of users associated with sports applications, and each user type will have their own set of tasks and



requirements [1]. As an example, research has found that coaches and athletes use data in different ways, and for different purposes [5]. Thus, a sports data visualization system will need to be designed specifically for its intended target sport.

2.8 Ethical aspects

Sports data is different from other private enterprises. Personal data about performance, injuries, and other information is tracked and measured constantly and becomes a point of discussion in contract- and salary-negotiations. Researchers who use sports data must therefore consider the risks and benefits, not only for management (teams and/or organizations), but for the labor-side (i.e. the athletes in this case) as well [11]. While these considerations are left to the discretion of the analysts, coaches and/or athletes that utilize visualization tools, it is still a relevant aspect to keep in mind for designers.

3. METHOD & RESULTS

As previously mentioned, this project followed a process of continuous feedback from domain experts inspired by the problem-driven approach for design study methodology proposed by Sedlmair et al., consisting of a nine-stage framework with three categories [14]. Since it is common to continuously move back and forth between stages and categories as part of an iterative process, methods of research, development and evaluation were divided into sub-sections corresponding to project activities. The process, illustrated in Figure 1, is explained throughout this section. As seen in the figure, different activities overlapped and affected each other in

various ways. The activities have been divided into two categories, *information gathering* and *planning, evaluation and implementation*. As seen in the figure, activities in the latter category used information from and helped generate new activities in the former category. The category called *artifact* consists of the prototypes designed, analyzed and discussed in this project.

The qualitative information gathered in this project was analyzed using thematic analysis. It is a method for identifying patterns of meaning (themes) across a dataset [3], where a theme “captures something important about the data in relation to the research question, and represents some level of patterned response or meaning” [2]. In order to discuss the results in relation to the research question, inspiration was taken from a number of evaluation scenarios for visualization design: A paper by Lam et al. discusses a number of evaluation scenarios which are relevant to this case, specifically *visual data analysis and reasoning*, *communication through visualization*, *collaborative data analysis*, *user performance* and *user experience* [6]. Furthermore, Sedlmair et al. list a number of possible contributions of a design study interesting to this case, especially *problem characterization and abstraction* and a *validated visualization design* [14]. Therefore, these are used as discussion topics in Section 4.

3.1 Pre-study

This first phase of the project was closely associated with the *precondition* phase of the Sedlmair et al. framework [14]. It consisted of planning, obtaining a theoretical ground for the project, as well as setting up

collaborations for the project going forward. First and foremost, this entailed literature study of topics related to the project, much of which is described in section 2, *Theory and Related Work*. Furthermore, collaborations were set up with *domain experts* from the SOC and SOA that would provide feedback throughout the iterations of this project.

3.1.1 Domain experts

Four people that for different reasons could provide expertise and insight into the area of data analysis for the athletic activities of the SOC were selected as collaborators for this project. Sedlmair et al. recognizes two critical roles for collaborators in a design study: *front-line analysts*, the domain experts performing data analysis, and *gatekeepers*, who have the power to approve or block the project. Other roles of interest are the *connector*, a person who connects the researcher to other interesting people, and the *translator*, who can abstract domain problems into more generic form, as well as relate them to larger-context domain goals [14]. Since the process was iterative, it was deemed reasonable to continuously work with this group, whose expertise over time could be combined with a familiarity with the concept to produce qualitative results. For brevity, they will be referred to as E1, E2, E3 and E4 or collectively as *the domain experts* in this report.

- **E1:** A researcher from the SOA, who was also the principal contact at the SOA for this project. They are responsible for the project from the SOA's side and therefore mediated all contact between stakeholders for this project. E1 functioned as a connector, a gatekeeper and a translator.
- **E2:** Responsible for performance analysis at the SOC. Bringing both the expertise of an analyst and the experience of being a former professional athlete, E2 provided the perspective of a front-line analyst and a translator.
- **E3:** Coach of the Swedish national kayaking team (also an analyst and former kayaking athlete). E3 had the role of both front-line analyst and translator, but also as gatekeeper due to their insight into kayaking, specifically.

- **E4:** As General Manager of the SOC, as well as having many years of experience in competitive sports, E4 had the roles of front-line analyst and translator for this project.

As a first information gathering activity, a focus group was held with the domain experts.

3.1.2 Initial focus group

This activity was aimed at gathering information on users and requirements for the tool through group discussion. Information was acquired through thematic qualitative analysis of a number of predefined themes, formulated as discussion questions during the focus group.

Discussion questions

- Who would use this tool (athletes, trainers, etc.)?
- Who would *not* use this tool (who might one think would be interested, but would in fact not)?
- When would the tool be used (before/after/during training)?
- When would the tool *not* be used?
- What are the most important features; what do users need to be able to do?
- Which functionality is required at the minimum for the tool to be of any use at all?
- What do you think is missing from the current methods and tools used for analysis, and what are these tools?
- What are some of the most interesting conclusions or insights that can be drawn from the currently available data?
- What is the end goal of using such a tool?
- Are there limitations as to what can be achieved through using the tool?

Predefined themes

- Who are the users?
- What are the requirements?
- What are the user tasks?
- When is the tool primarily used?
- What can be achieved using an analysis tool (what are the goals)?
- What can perhaps *not* be achieved using an analysis tool?

Based on the domain experts opinions on these topics, the following conclusions could be drawn. The users can be divided into different categories, which all have different requirements for the visualization tool. For the purposes of this project, they are defined as follows:

- *Athletes* - currently active kayaking athletes working with the SOC and/or SOA. Will often prefer video documentation to raw numbers. However, it is important to note that this is not true for all athletes, as they will have varying interests in analytics. Generally, they will want concrete evidence that insights from data analysis will lead to better performance.
- *Coaches* - responsible for improving the overall performance of an athlete. They assess the athlete's actions, either by direct observation, reviewing video footage or analyzing collected training data. A coach will use their knowledge of the sport to draw conclusions, sometimes supported by data, and will often contact a performance analyst with requests for looking at specific situations. The coach works as a bridge connecting the analyst and the athlete, in many cases.
- *Performance analysts* - the analysts will perhaps be the most frequent users of the visualization tool. Their primary duty is to analyze athlete's performances. Currently, this is achieved through video footage or raw numbers that can be manually filtered and presented as graphs, or simply as numbers in a table. They will often work together with a coach, looking at a specific situation. In reality, the analyst many times works as a bridge between data and *both* the athlete and coach, simultaneously.
- *Trainers* - has a specific area of responsibility and expertise, e.g. strength. Their work is important for improving performance and preventing injury. A coach and/or analyst can use gathered information (video, qualitative data) to get feedback from the trainer.

With regards to user tasks, a number of important implications were discovered. The tool needs to

include a variety of variables, including (but not limited to) acceleration, rotation, speed, pulse (heart rate), time and deltaV (speed changes when paddling). Much of the analysis is centered around comparisons, finding relationships, correlations and outliers in the data. A system that enables exploratory tasks was an apparent need expressed by the domain experts. There was also a request to analyze specific phases, such as the start phase of a race, as well as how the boat moves (i.e. tilts or rotates) and how that correlates with other variables.

3.2 Conceptual study

3.2.1 Concept

The project shifted to the next phase, creating a concept for the proposed visualization tool. As illustrated in Figure 1, this stage involved combining information acquired during both activities of the pre-study phase. This phase consisted of the early steps from the *core* phase of the Sedlmair et al. framework [14]. The objective was to characterize and abstract the problem at hand, and arrive at a shared understanding of the problem with the domain experts. Briefly summarized, the goal was a web-based tool that allows athletes, coaches, analysts and trainers to generate insights about training data, either alone or in collaboration with each other. The tool should be easy to use, allowing users to interact and shape their own discovery process.

3.2.2 Low-fidelity prototype

The concept was formulated as a number of design hypotheses. These were developed as digital sketches using the prototyping software Sketch, resulting in a *Low-fidelity prototype*. It consisted of six visualization concepts and four concepts regarding more meta-oriented functionality, such as data management. It was used to generate feedback from the domain experts on the concept at this stage of the process. Examples of some of the visualization views are shown below.

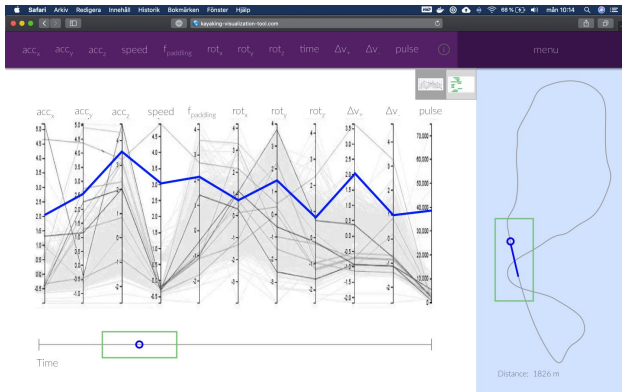


Figure 2. The *compare all* view, showing all variables in parallel coordinates. Note the *boat route* to the right.

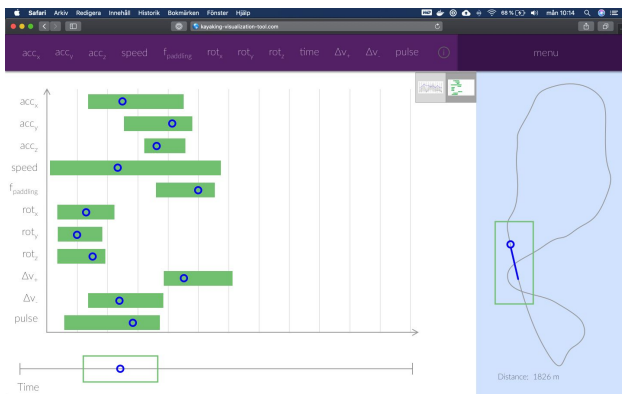


Figure 3. An alternative to the view shown in Figure 2, with a *span chart* instead of the *parallel coordinates*.

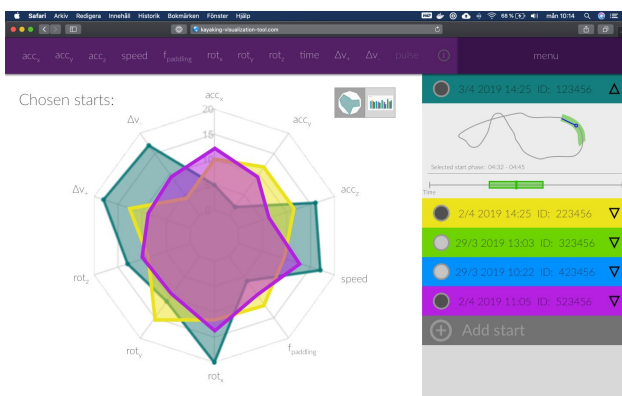


Figure 4. The *start analysis* view, showing a number of data subsets and comparing them using a radar chart.

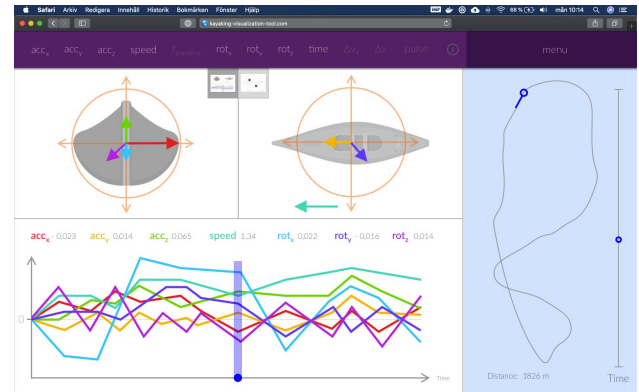


Figure 5. The *boat movement* view, showing acceleration and rotation of the boat, with the rest of the chosen variables displayed in a line chart.

3.2.3 Feedback 1

A discussion was held over video conference with E1 and E3 to provide feedback on the design proposals. The sketches were also sent to E2 and E4 to provide context for subsequent evaluation activities. E1 and E3 were shown all of the design hypotheses, and were asked to discuss all of the design elements for each of the visualization views. Their feedback was analyzed using the following predefined themes:

- Overall good feedback or usefulness
- Very good feedback or usefulness
- Possible improvement and/or new idea
- Caution, visualization might be misleading
- Not useful or something that needs to be fixed

The *boat route* (Figures 2, 3, 4 and 5) was deemed a very useful feature. The *span chart* (Figure 3), used as an alternative view to the more detailed *parallel coordinates* (Figure 2), was deemed more understandable, especially to a user with less interest in numbers and analysis. Furthermore, E1 emphasized the importance of being able to filter out uninteresting variables and to be able to rearrange them in the visualization, wherever possible. The experts also requested a way to “play” through the data, point by point to see how it changes over a specified time period. The *start analysis* view (Figure 4) was deemed useful, but needed a few improvements. For instance, only displaying single values from a specific point in time introduces a danger of misrepresenting the data. Displaying the rest of the data in a specified

time span to provide context would be a way of mitigating this problem, according to the domain experts. Another view that was deemed promising, but had some issues in its current state, was the *boat movement* view (Figure 5). The rotation should not be represented by arrows, as this might be confusing. Furthermore, the *front view* should instead be changed to a *back view* to better represent the boat moving forward. Finally, the image used for the kayak should be changed to one of a racing kayak, as the current one resembled a recreational boat. Using an incorrect image would negatively impact the credibility of the tool.

3.3 Design study

The feedback from the activity described in section 3.2.3 was analysed and provided background for the next phase of the project: Realizing the sketches into a working high-fidelity prototype. This phase is what Sedlmair et al. describe as the interleaved *design* and *implement* stages of the *core* phase [14].

3.3.1 Implementation

Four of the design concepts evaluated in the previous phase were implemented, focusing primarily on the visualization views. Some of these views were made up of two alternatives, called *subviews*. As illustrated in Figure 1, this stage required data from all former information gathering activities. The views that had been deemed the most interesting or useful by the domain experts in the previous phase were prioritized in the implementation of the high-fidelity prototype. A sample dataset from a kayaking training session was provided by E1.

3.3.2 High-fidelity prototype

This section will detail the implemented visualization tool that was tested by the domain experts. The sample dataset, some common design elements and each of the four implemented views are described below:

Sample dataset

The dataset consisted of 12 variables: speed, time, deltaV (two variables, negative/positive speed

change), coordinates (longitude and latitude), acceleration and rotation (both in the x-, y- and z-directions). Data for pulse (heart rate) was requested by the domain experts but was not available in the sample dataset, and therefore had to be generated as mock data. Thus, there were 13 variables available for the tool, each with 8 529 data points.

Common design elements

There were three design elements visible on all of the visualization views:



Figure 6. The timeline with two play buttons.

The timeline (Figure 6) was a way for users to narrow down the selection of data points for each view using three handles. The middle handle on the timeline defines a single point within the selection to be highlighted. There were also two *play* functions:

- *Span* moved the middle handle along the timeline for each point in the selection, thus seeing how the data varies over that time.
- *All* worked in a similar way, but instead moved the width of the selection over the entire timeline.



Figure 7. The variable menu.

The top menu (Figure 7), provided a way for users to filter out uninteresting variables. The selection also persisted in the other views.

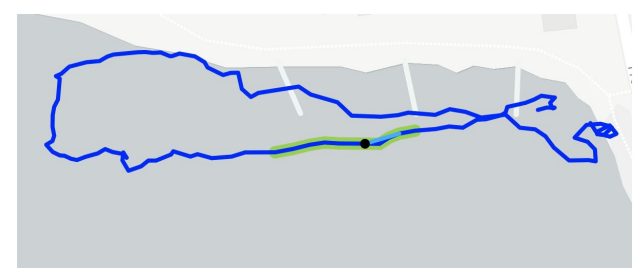


Figure 8. A map with the boats' travelled route plotted.

There was also a map with a route plotted from the coordinates in the sample dataset (Figure 8). The currently chosen point on the timeline is shown by a black dot, followed by a turquoise line representing the speed at that point. Furthermore, the span defined

using the timeline (Figure 6) is highlighted in green on the plotted route.

Compare all variables

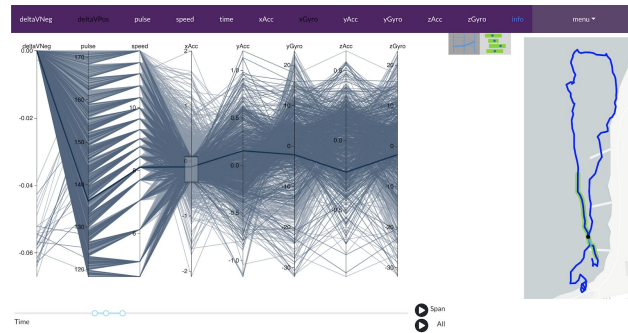


Figure 9. The primary subview of the *Compare all variables* view.

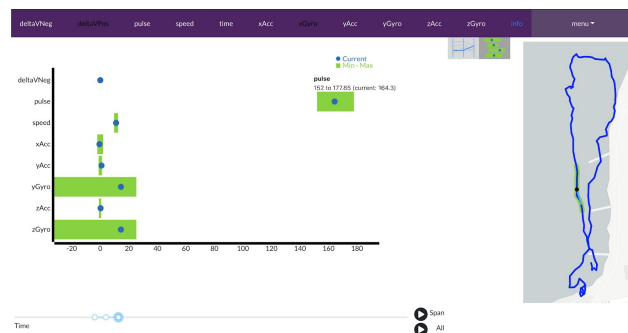


Figure 10. The alternative subview of the *Compare all variables* view.

The *Compare all variables* view (Figures 9 and 10) let users explore relationships between variables. The primary subview (Figure 9) consisted of a parallel coordinates plot showing all points in the current selection. The currently selected point is shown by a bold line. All variable axes can be rearranged, and users can brush axes to highlight points within a specified range of values (Figure 9). The subview with the span chart (Figure 10) was an alternative to the parallel coordinates, showing the max and min values in the selected span of each variable, with the current value shown as a blue dot. If the user hovered over a bar, the exact values were shown in text.

Speed comparison



Figure 11. The *Speed comparison* view.

The *Speed comparison* view (Figure 11) showed how the speed variable relates to all other variables. A line chart was used to show the speed over the entire dataset. For the rest of the variables, a box-and-whiskers plot was used. The whiskers represented the maximum and minimum values of that variable within the entire dataset, the height of the box was the maximum and minimum for the selected time span, a line on the box showed the median value for the selection and a circle on the box showed the currently selected point. Hovering over a box showed an information window with all values.

Start analysis

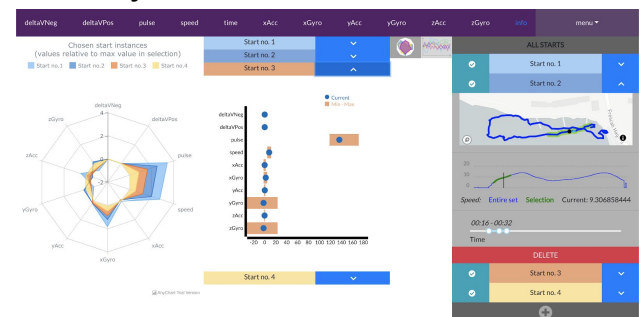


Figure 12. The primary subview of the *Start analysis* view.

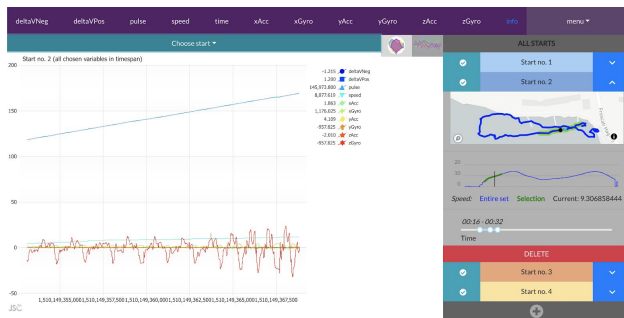


Figure 13. The alternative subview of the *Start analysis* view.

The *Start analysis* view (Figures 12 and 13) provided a way for users to analyze and compare specific events of a training session. Users could add new entities using the menu shown to the right, specifying the selection and a point of interest using the timeline sliders. To navigate the dataset properly, a line chart showing the speed and selection was available. The selections were compared using a radar chart (Figure 12), showing the currently selected point in relation to the maximum value for each variable in the selected time span. There was also a details view with a span chart for each selection (Figure 12). Furthermore, there was an alternative subview shown in Figure 13 with a line chart for each selection, showing all active variables in the specified time span.

Boat movement

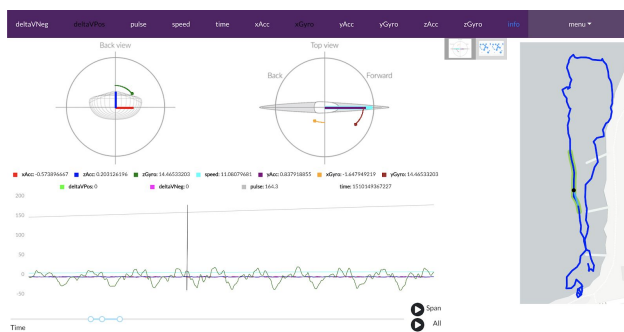


Figure 14. The primary subview of the *Boat movement* view.

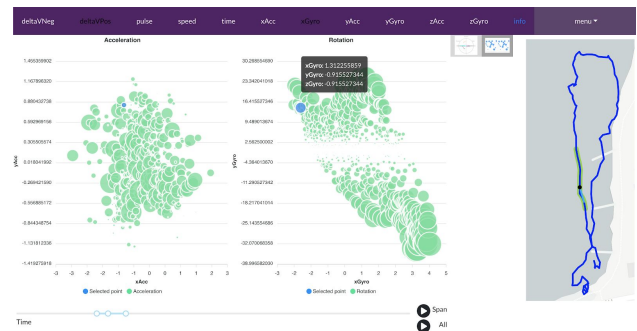


Figure 15. The alternative subview of the *Boat movement* view.

The final view in the tool was *Boat movement* (Figures 14 and 15), focused on how the acceleration, rotation and speed of the boat varied. The primary subview (Figure 14) showed two images of a boat with colored lines and curves representing the different variables. All values for the currently selected point were shown in a legend below the images, and at the bottom there was also a line chart showing all active variables over the selected time span. There was also an alternative subview (Figure 15) where acceleration and rotation was shown separately in two bubble charts. In each of these cases, the x- and y-variables were represented by their respective axes and the z-variable decided the size of the bubble. The selected point was shown in a different color. Hovering over one of the bubbles showed an information window with the values for that point.

3.3.3 Feedback 2

Meetings were held over video conference, with the domain experts testing the tool on their computer and sharing their screen with the researcher. Three tests were held: The first test consisted of showing the tool to E1, without them interacting with the tool. Test 2 was held with E2 and E4 simultaneously: E4 interacted with the tool on their computer while E2 watched, both engaged in the discussion. The final test was held with E3, who interacted with the tool and provided feedback. No specific information gathering tasks were given, as the experts were simply asked to try all of the functions detailed in Section 3.3.2. The experts were asked to give their own

opinions first, and then discuss from a predefined set of topics after testing/seeing a certain part of the tool before moving on. As such, the results of the tests were a combination of discussions and researcher observations. The results of the tests were analyzed thematically.

For each test, the views were tested in a different order, and the experts were asked to compare the current view to any previous views. Furthermore, they were all asked specific questions for each view:

Compare all variables

- Which of the two subviews did you prefer?
- Did you understand what information was being displayed?
- Was the play function useful in this view?

Speed comparison

- Did you understand what the box-and-whiskers plot was meant to show?
- Was the play function useful in this view?

Start analysis

- Did you understand what the radar chart was meant to show?
- Were the two details-views (span chart and line chart) useful?

Boat movement

- Do you consider the directions shown in the visualization to be realistic?
- Was the line chart useful?
- Was the play function useful in this view?
- Which of the two subviews did you prefer?

After testing *each* view, the experts were also asked to discuss the following topics:

- Whether they felt that any functionality was missing from the tool, or if they identified specific areas of improvement.
- How easy it was to explore information in this particular view.
- Whether the tool is usable and/or useful.
- Whether the tool is clear with regards to what information is being displayed.

- Whether the tool can be learned by using it.

Finally, after seeing *all* of the views, the experts were asked to discuss the following topics.

- How this experience differs from their experience of current performance analysis situations.
- Whether any specific part of the tool stood out to them.
- Whether the tool would help to make data analysis for sports easier.
- Whether new information can be generated by using the tool.
- Whether they think it is possible to come up with new theories by using the tool.
- Whether the tool would make the communication of ideas with colleagues easier.
- Whether it is likely that the tool could help decision making (e.g. about future training activities).
- If there is anything else they would like to add.

3.3.4 Feedback 2 - Results

The tool received overall positive feedback from the experts, but with implications that several improvements are needed in further development.

Technical issues

The tests being conducted over video conferences affected the bandwidth, resulting in increased loading times and the tool responding slower to user input. However, there were also issues with the tool performing slowly and not being able to load all data points when using the play-functions described in section 3.3.2. Furthermore, there were some browser-specific problems that arose during testing as well, when the tool was used on the domain experts' own computers.

Accounts of the tool being intuitive and/or useful

Several instances of the experts being able to intuitively understand and explore the tool without instruction were noticed during testing. Furthermore, some of the functions in the tool received positive feedback from the experts in this regard. Visualizing the boat route using the map (Figure 8) was deemed helpful, providing a level understanding and context. For the *boat movement* view (Figure 14) most of the directions shown by the lines and curves were

deemed realistic and understandable. The span chart used in the *start analysis* (Figure 12) and *compare all* (Figure 10) views was deemed to be very intuitive and understandable by the domain experts.

Moreover, the play functionality described in section 3.3.2 was considered a good way of viewing how data changed over the selected time period. The brushing and rearranging functionality in the parallel coordinates plot also received praise from the experts. Also, using the variable menu (Figure 7) was helpful in letting the domain experts focus on what they wanted to see. Using a line chart in the *start analysis* view (Figures 12 and 13) was also deemed helpful, both as a way of providing details for variables of interest, and for identifying starts in the dataset based on how the speed varied.

Accounts of the tool *not* being intuitive

The parallel coordinates plot (Figure 9) had several issues with intuitiveness: 1) The bold line representing the chosen data point was initially misinterpreted; 2) The availability of the brushing functionality was not apparent to the experts; 3) Overplotting was reported by all experts, due to the large number of available data points, even after narrowing data selections. Furthermore, the box-and-whiskers plot (Figure 11) lacked clarity in terms of what was being displayed, despite having a details-tooltip. The domain experts expressed a need for instructions or some sort of manual in these instances.

There were also issues with interference for data with different value ranges: For instance, the *pulse* variable in the test data set ranged from 65 to 195, while the *deltaVpos* variable ranged from 0 to 0.68. When shown in the same plot, for example a line or span chart, this caused the *deltaVpos* values to be all but hidden from sight.

Missing functionality

The domain experts mentioned certain things that they felt were missing from the tool:

- Zooming functionality for the timeline to allow more precise data selections.

- Comparing different datasets, e.g. for several training sessions, multiple athletes in relation to each other, or providing a set of reference data.
- A standard set of selected variables for each view.
- The map should have the total distance travelled, as this information is valuable to analysts.
- A way to identify the max value in a given selection and explore that part of the dataset.
- A way to save a comment along with the data.

Possible improvements

Some specific improvements were suggested for various parts of the tool:

- There were several suggestions for the *boat movement* view (Figure 14) to improve clarity:
 - Make the images move, instead of the lines and curves.
 - Create a 3D-model of a boat that moves according to the appropriate values.
 - Add a third boat-image (from the side) to better show the x-rotation and z-acceleration.
- Selecting subsets of the data based on distance or speed rather than time, resembling how they currently identify segments of interest for analysis.
- The line chart used as a details-subview for the *start analysis* view (Figure 13) could be improved by instead showing all of the defined starts, but with one variable selected at a time.
- The *speed comparison* view (Figure 11) could be changed to allow for comparing *any* single variable to all others instead, to extend its area of applicability.
- E1 suggested limiting the number of decimals shown, to improve overall readability.

Need for clearly defined tasks and/or goals

This was expressed in different contexts throughout the tests. Partly as a way to mitigate what was an otherwise potentially confusing view, like the parallel coordinates (Figure 9), but also overall in the tool. The domain experts pointed out the importance of defining visualization and analytical tasks by first having observed something during training, that could then

be further examined and possibly explained by using the tool. Also, having a clearly defined goal was deemed important by the experts, as they felt that data analysis was not an end goal in and of itself. The experts agreed that such tasks and goals would affect the perceived ease with which information can be discovered by using the tool.

Other insights from domain experts

One insight that has been expressed by the domain experts throughout the process, is the fact that different users of this tool will have different needs and prerequisites. It will depend on the users role within the SOC, but will also differ from person to person, as the predisposition for data analysis will sometimes be a matter of personal preference; some will think of it as a fun and motivating exercise, while some will instead want to use more practical methods.

The *start analysis* in general and the radar chart (Figure 12) specifically was deemed to enable meaningful comparisons, and that principle could possibly be applied to other parts of the tool and to other training activities as well, according to the experts.

Furthermore, when asked whether the tool would make data analysis *simpler*, E2 stated that this term may be misleading. Instead, they proposed that the tool might help to make data analysis for sports more *sophisticated*, providing deeper and more thorough analysis, rather than simplifying what is currently done in that area.

Some differences from the current analytical methods were also identified. For instance, some of the elements were deemed to be “more graphical” than some of the currently used tools, which E3 claimed made it easier to learn. Furthermore, using imagery and design elements that allude to the sport in question was welcomed by the experts, as it provided context and gave the tool credibility. Also, familiarity with certain visual mappings (such as line, bar or radar charts) resulted in a higher level of understanding amongst the experts.

Data management suggestions

The segmentation of datasets into smaller parts, which was the primary idea behind the *start analysis* view (Figures 12 and 13) was deemed a positive

element for analysis overall, and can be used to a greater extent. A way of splitting datasets into parts was therefore requested by the experts. Furthermore, it was expressed that there should be a coherent and unified way of collecting data for the SOC.

Perceived benefits of the tool

There seemed to be support for the idea that the tool would be able to help with the generation of new knowledge, and that it could be used to create new methods or alternative ways of working within the sport. Furthermore, the domain experts thought that the tool would be able to support decision making and help substantiate a user’s arguments in collaboration with colleagues. Specifically, this could aid coaches when attempting to convince an athlete that a change of some sort should be implemented. E2 also added that if the tool is successful, data analysis could prove to become a valuable alternative training activity for athletes, especially late in their career.

4. DISCUSSION

This was the final stage of the process, where information from all activities was gathered and discussed. This is illustrated as the *Analysis* phase on the process shown in Figure 1. In the Sedlmair et al. framework, the *analysis* phase is defined by the *reflect* and *write* stages and is focused on justifying the results of a design study to the outside world [14]. The topics discussed in this section were mainly inspired by the scenarios listed by Lam et al. [6] and by the possible contributions for design studies listed by Sedlmair et al. [14] (mentioned in Section 3).

4.1 Problem characterization and abstraction

This topic is described by Sedlmair et al. as a “first-class” contribution of a design study [14]. They claim it to be important for several reasons, first and foremost for achieving understanding between the domain experts and the researcher. For this case specifically, the overarching problem was fairly well formulated from the beginning: The SOA wanted an interactive visualization tool for kayaking training data. However, as the process went on, the problem was abstracted and divided into smaller, more well-defined subproblems.

These subproblems are probably best illustrated by the different visualization views suggested in this project, and new ones will likely arise in any future work. Improvements and new features for achieving different goals were expressed by the domain experts in all feedback sessions. It is therefore possible that this problem, due to being exploratory and ill-defined in nature, will require developing and evaluating many more solutions in future work.

An interesting topic is the different user types with different tasks and requirements, which was discussed early on and returned continuously throughout the process. As mentioned in section 2.7, this sentiment has been stated for visualization applications in sports elsewhere as well [1, 5]. Since the interpretation of a visualization is an internal process [8], each user's own predisposition will influence their use of the tool. It has also been argued that adaptive and exploratory data analysis, which represents the tasks performed with this tool, will be influenced by previous conclusions and insights [4]. This is important to consider in any future work.

4.2 Visual data analysis and reasoning

While the tool is not finished in its current state, the domain experts seemed to think that there was potential for data exploration and information seeking in the tool. They also expressed that the goal of seeking evidence for or against a self-formulated theory would also be possible through the tool. This confirms the notion of interactive visualization as an effective way of generating contextual insights [13].

Specific functions that were considered useful often enabled filtering out uninteresting information. This is coherent with the *visual information-seeking mantra* mentioned in section 2.3, specifically the step *zooming and filtering* [16]. Meanwhile, the domain experts wanted to introduce functions solely focused on “playing around”, indicating that the design could be improved with regards to exploration. The domain experts highlighted the abilities for meaningful comparisons displayed in the *start analysis* view (Figure 12), and thought it showed promise for other areas as well. This could implicate that the concept enables

generating complex insights [10], and could be explored and evaluated further.

4.3 User experience and performance

As detailed in Section 3.3.4, there were features of the tool that were deemed useful, functionality that was missing, and features that could be reworked and/or improved.

Some missing functionality was identified for the tool, perhaps most obviously a lack of instructions. While this was a conscious choice to better understand the intuitiveness of certain visual mappings, there is likely much that can be achieved by exploring this concept further. This could make the tool more accessible, and thereby used by more people within the SOC. Meanwhile, E3 claimed that the tool was “more graphical”, and therefore easier to learn than those currently used for analysis. This could indicate that the tool would improve current methods, providing more intuitive tools for analysis. Also, it goes along with related research stating that visualization can be more effective for insight generation than traditional statistical analysis [18].

The data being characteristically diverse lead to problems with visibility, which is mentioned in Section 3.3.4. In these cases, solutions like separating variables with similar value ranges into different plots could be considered. Alternatively, the values could be presented in relation to their respective value range, as was the case with the *start view* (Figure 12). However, this must be done carefully to not risk misrepresenting the data. Alternatively, data specification and selection could be re-done together with the domain experts, to enable the suggestion of alternative visualizations [7]. The parallel coordinates view had issues with overplotting, due to the large number of data points. It is possible, however, that there are other uses for this visual mapping, perhaps to show several start instances together and compare specific variables. Meanwhile, the large number of data points affected the usability of the timeline as well, necessitating adding features to that function.

Comparing the different views to each other, it is evident that using visual mappings familiar to the domain experts made views more intuitive. However, the domain experts fairly quickly learned to interact with mappings such as the parallel coordinates or box-and-whiskers plot. Combined with the expressed need for guidance or instructions in the tool, it seems likely that unfamiliar visual mappings can be useful as well. The domain experts seemed to agree that the tool was indeed understandable and could be learned. An interesting point is that imagery that alluded back to the sport in question was perceived to improve usability, lend credibility and provide context.

4.4 Communication and collaboration

Generating new knowledge about the sport seemed possible with this tool. The notion that this could lead to new methods and alternative ways of working in the sport is also promising for the concept. The final sentiment from the results in section 3.3.4 is especially interesting, regarding the use of the analytics tool as a training activity in and of itself. Furthermore, since previous experience is important for extending knowledge of an area, it is possible that combining the experience and knowledge of several users will improve collaborative analysis [12].

Furthermore, the domain experts argued that communicating insights is made possible through the tool, and that it could help motivate changes in training exercises for athletes, for example. According to the domain experts, this is intertwined with the aforementioned goal of proving the theories of a coach, in order to convince a trainer or athlete. If successful, this could implicate that the tool has a high level of efficacy, since it helps the expert achieve a professional goal [15]. Furthermore, these topics are highlighted as important roles for visualization systems by related research, and fulfilling them would thus lend credibility to this concept [17].

4.5 Validation of the visualization design

According to the feedback from the domain experts, the design is certainly not finished at this point. This is evident from the issues mentioned in subsections *missing functionality* and *possible improvements* of Section 3.3.4. However, due to the positive feedback

for the concept as a whole, it is possible that this could be achieved through further research and development. The domain experts expressed opinions that the tool would both help make the analysis process easier *and* provide a higher level of sophistication. This could be interpreted as the *concept* showing promise, even if the design is not fully completed. However, the tool needs further development, implementation and evaluation.

4.6 Future work

Apart from what has already been mentioned in this section, future research and development of this project has a number of interesting areas to work with. Some specific suggestions from the domain experts are detailed in *missing functionality* and *possible improvements* of Section 3.3.4.

Some of the technical issues referenced in Section 3.3.4 could be attributed to the tool itself, which is something that should be considered moving forward. The tool performing slowly and responding poorly to user input will negatively affect the user experience. As has previously been stated, performance is a key factor for the success of an interactive visualization tool [16, 9]. Therefore, future work on this design should include performance optimization.

Data management should be investigated further, due to the domain experts expressing the need for further segmenting the dataset and introducing comparisons to other datasets as well. There was also a request to be able to record subjective comments as part of the analysis tool. This qualitative data could be combined with information about the athletes to further research the concept of *meta-data* for this project [11].

Finally, the possibility of extending the use of this tool to other sports is something that should be considered as well. This was expressed by the domain experts, and sports remains an interesting area to explore with regards to visualization. Because of the need for designs specific to the sport in question, it is possible that novel insights can be found to further this research area [1].

4.7 Method criticism

As previously mentioned, this project followed a process inspired by the problem-driven approach proposed by Sedlmair et al. [14]. The results cited as common contributions for this methodology is *problem characterization and abstraction*, a *validated visualization design*, and *reflection*. While these are discussed in this section, it is possible that even stronger contribution claims could have been made, had the concept developed in this project been fully implemented and evaluated.

Furthermore, all data collected in this project was analyzed using thematic analysis, which seems appropriate given that all of the information was of a qualitative nature [2, 3]. If the concept described in this project were to be developed and implemented further, methods of including and evaluating quantitative data should also be considered.

Finally, the possibilities for user testing was impacted by the outbreak of COVID-19, which occurred at the early stages of the project. Specifically, some technical issues described in Section 3.3.4 can be attributed to this fact, since this forced the tests to be carried out over video conference. The same can be said with regards to the *Feedback 1* activity described in Section 3.2.3, where not all domain experts could attend the meeting.

5. CONCLUSION

This report details the conceptual design of a visualization tool for kayaking. In this concept, interactive visualizations provide opportunities for experts with various roles within the sports organization (athletes, coaches, analysts) to generate advanced insights through exploratory data analysis. The wide range of available data in a multidimensional dataset enables finding connections that would not otherwise be obvious. Using the tool could be a way to improve the quality of the analysis, and make the process easier and more accessible. Furthermore, it allows for combining the experience and knowledge of several users to improve collaboration. The tool

was found to be intuitive and easy to learn, and using sport-specific imagery reportedly improved the user experience.

There are certainly challenges associated with needing to design the concept for a specific sport, since this can affect its transferability to other sports. Also, everyone in the sports organization will not be interested in quantitative data analysis, opting for more traditional methods instead. This necessitates the development of a wide range of features, to satisfy different user needs. The development of such a tool requires expert knowledge of the sport (and organization) at hand. Also, using multidimensional datasets with different data characteristics and value ranges, entails challenges when this data is visualized together. This emphasizes the need for a close and continuous collaboration with domain experts as part of the process.

With regards to problem characterization and abstraction to this specific case, important insights found early on were reformulated and deepened throughout an iterative process. A need for certain solutions was found and formulated, and the concept showed promise in testing, but further work is needed. Thus, the problem cannot be considered solved, but promising steps have been taken for future work.

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